

Supporting Information for “What Must a Pore-Scale Transport Model Remember? Memory Adequacy in Non-Fickian Transport”

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Purpose and Reproducibility

This supporting information records the numerical details behind the memory-adequacy results reported in the main text. The main manuscript keeps the scientific design compact: a resolved trajectory archive, competing memory mechanisms, and held-out observables. The tables below provide the audit trail for that design, including the tight OpenFOAM resolution ladder, particle-tracking tolerances, objective-weight sensitivity, and the evidence matrix used to construct the memory-adequacy atlas.

The OpenFOAM cases use voxel finite-volume meshes extracted from the same Core2 Ben-theimer subvolume. The 18 μm and 12 μm cases use downsampled voxel meshes, while the 6 μm case uses the full image-resolution voxel mesh and a strict SIMPLE solve with zero linear-solver relative tolerances and residual-control thresholds of 10^{-9} for pressure and velocity. The trajectory archives used for the manuscript-facing OpenFOAM tests contain 5000 particles, 4000 saved steps, and saved-step interval $\Delta t = 0.1$. Particle integration used internal substepping with physical tolerances scaled across the resolution ladder; no case reached the substep cap.

Solver and Tracking Audit

Table S1: OpenFOAM resolution-ladder and particle-tracking audit for Core2. The 6 μm case is the strict full-resolution solve; the listed residual is the largest final OpenFOAM initial residual among pressure and velocity components at convergence. Lower-resolution residuals were not rerun with the same strict residual audit, so the table emphasizes their accepted flow fields and particle-integration diagnostics.

Voxel size	Cells	Flow time	k (m^2)	Max residual	Particles	Max substeps	Cap hits
18 μm	98,270	103	4.277×10^{-12}	—	5000	4	0
12 μm	322,524	105	3.690×10^{-12}	—	5000	5	0
6 μm	2,650,688	604	3.492×10^{-12}	9.94×10^{-10}	5000	9	0

Supporting Tables

Table S2: Tight particle-tracking tolerances across the OpenFOAM resolution ladder. The tolerance scaling preserves physical distance rather than voxel count. Mean substeps are reported per saved output step.

Voxel size	D	Max adv. step	Max diff. step	Mean length	Mean substeps
18 μm	10^{-3}	0.0333	0.0250	4001.000	1.0086
12 μm	2.25×10^{-3}	0.0500	0.0375	4000.842	1.0108
6 μm	9×10^{-3}	0.1000	0.0750	4000.689	1.0101

Table S3: Outer-split robustness on Core1 baseline trajectories. Lower objective and rank are better.

Mechanism	Mean obj.	Std. obj.	Mean rank	Wins	Beats GB/H
Validation mixture	121.64	47.75	2.00	2	3/3
Gaussian/Bayes	125.48	50.22	2.00	2	-/4
hybrid	127.71	57.25	3.20	1	1/-
Bootstrap mixture	132.05	50.44	3.20	0	1/3
kNN conditional	142.47	38.49	4.80	0	0/1
pair-rerank	167.72	60.16	5.80	0	0/0

Table S4: Best mean mechanism under different Core1 objective-weight regimes.

Regime	Best mean mechanism	Mean obj.	Mean rank
Balanced	Gaussian/Bayes	145.26	2.25
BTC heavy	Gaussian/Bayes	188.73	2.25
Pair heavy	kNN conditional	204.41	2.75
Dilution heavy	hybrid	170.07	2.50
Reaction light	Gaussian/Bayes	140.53	2.25
Reaction heavy	Gaussian/Bayes	85.75	1.75
No reaction	Gaussian/Bayes	140.01	2.00

Table S5: Mean selected mixture weights across the Core1 Peclet sweep. Weights are ordered as velocity memory, archive proximity, learned context, and pair organization.

Condition	Gaussian/Bayes	kNN	hybrid	pair-rerank
$D = 3 \times 10^{-4}$	0.4375	0.1875	0.3333	0.0417
$D = 10^{-3}$	0.4125	0.1500	0.4125	0.0250
$D = 3 \times 10^{-3}$	0.2708	0.0000	0.7083	0.0208

Table S6: Core2 18 μm OpenFOAM tight-archive outer-split mechanism performance. Lower objective and rank are better. The benchmark uses 5000 tracked particles and a strided archive of path motifs.

Mechanism	Mean obj.	Std. obj.	Mean rank	Wins
kNN conditional	1567.42	101.70	2.75	1
Validation mixture	1595.51	164.09	3.00	1
Gaussian/Bayes	1614.39	112.04	4.00	1
Bootstrap mixture	1619.47	87.60	3.00	1
pair-rerank	1650.90	57.19	4.25	0
hybrid	1657.55	16.51	4.00	0

Table S7: Best mean memory under different objective-weight regimes for the tight OpenFOAM resolution ladder. This table supports the main-text claim that the selected memory is conditional on resolution and observable: archive proximity is favored throughout the 18 μm case, validation-selected mixtures dominate the 12 μm case, and the 6 μm case preserves a velocity-memory exception for the pair-heavy objective.

Objective regime	18 μm	12 μm	6 μm
Balanced	Archive proximity	Validation mixture	Validation mixture
Breakthrough only	Archive proximity	Validation mixture	Validation mixture
BTC heavy	Archive proximity	Validation mixture	Validation mixture
Pair heavy	Archive proximity	Validation mixture	Velocity memory
Dilution heavy	Archive proximity	Validation mixture	Validation mixture
Reaction light	Archive proximity	Validation mixture	Validation mixture
Reaction heavy	Archive proximity	Validation mixture	Validation mixture
No reaction	Archive proximity	Validation mixture	Validation mixture

Table S8: Memory hypotheses evaluated in the main manuscript.

Memory hypothesis	Implementation	Primary information used	Role
Memory-free	Unconditional	Archive frequency only	Baseline forgetting
Archive proximity	kNN conditional	Local empirical proximity	Nonparametric matching
Velocity memory	Gaussian/Bayes	Diffusion-scaled velocity continuity	Original physics kernel
Learned context	hybrid	Contrastive learned score plus Gaussian/Bayes	Learned transition memory
Pair organization	pair-rerank	Pair descriptors plus Gaussian/Bayes	Short-horizon spatial organization
Memory mixture	Validation-selected mixture	Component weights selected by held-out score	Objective-aware memory selection

Table S9: Trajectory ensembles and benchmark roles.

Ensemble	Geometry	Velocity field	Diffusivity	Role
Core1 baseline	Core1	Graph-Laplace	10^{-3}	Baseline outer-split mixture benchmark
Core1 high Pe	Core1	Graph-Laplace	3×10^{-4}	Peclet-regime sensitivity
Core1 low Pe	Core1	Graph-Laplace	3×10^{-3}	Peclet-regime sensitivity
Core2 graph	Core2	Graph-Laplace	10^{-3}	Geometry transfer
Core2 OpenFOAM 18 μm	Core2	Finite volume	10^{-3}	Tight flow-fidelity test
Core2 OpenFOAM 12 μm	Core2	Finite volume	2.25×10^{-3}	Tight resolution-ladder test
Core2 OpenFOAM 6 μm	Core2	Finite volume	9×10^{-3}	Strict full-resolution flow test

Table S10: Evidence matrix across the benchmark conditions. Weights are ordered as velocity memory, archive proximity, learned context, and pair organization.

Condition	Best mean memory	Mean obj.	Mean rank	Selected weights
Core1 high Pe	Validation mixture	276.15	2.00	0.438, 0.188, 0.333, 0.042
Core1 baseline	Validation mixture	121.64	2.00	0.413, 0.150, 0.413, 0.025
Core1 low Pe	Learned context	243.53	2.50	0.271, 0.000, 0.708, 0.021
Core2 graph	Velocity memory	255.40	1.75	0.479, 0.188, 0.271, 0.063
Core2 OpenFOAM 18 μm	Archive proximity	1567.42	2.75	0.167, 0.417, 0.083, 0.333
Core2 OpenFOAM 12 μm	Validation mixture	1593.69	1.50	0.063, 0.479, 0.125, 0.333
Core2 OpenFOAM 6 μm	Validation mixture	1695.18	2.50	0.271, 0.417, 0.104, 0.208

Table S11: OpenFOAM resolution ladder for Core2 using tight 5000-particle trajectory archives. The autocorrelation values are axial velocity autocorrelations of the resolved reference trajectories at the listed saved-step lag with $\Delta t = 0.1$.

Voxel size	Cells	k (m^2)	Corr. lag 10	Corr. lag 80	Balanced best	Mean obj.
18 μm	98,270	4.277×10^{-12}	0.1099	0.0878	Archive proximity	1567.42
12 μm	322,524	3.690×10^{-12}	0.1259	0.1019	Validation mixture	1593.69
6 μm	2,650,688	3.492×10^{-12}	0.1260	0.1043	Validation mixture	1695.18